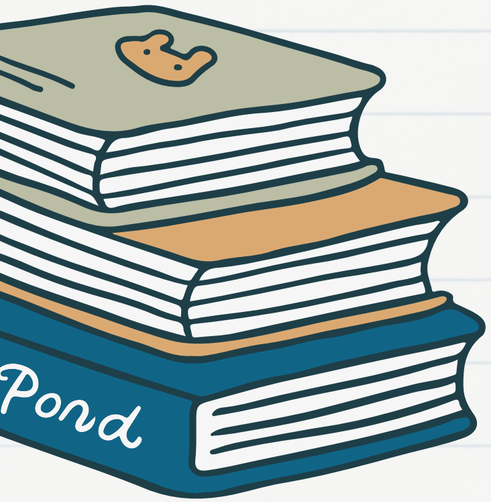


Object Oriented Concepts 2
Library Management System

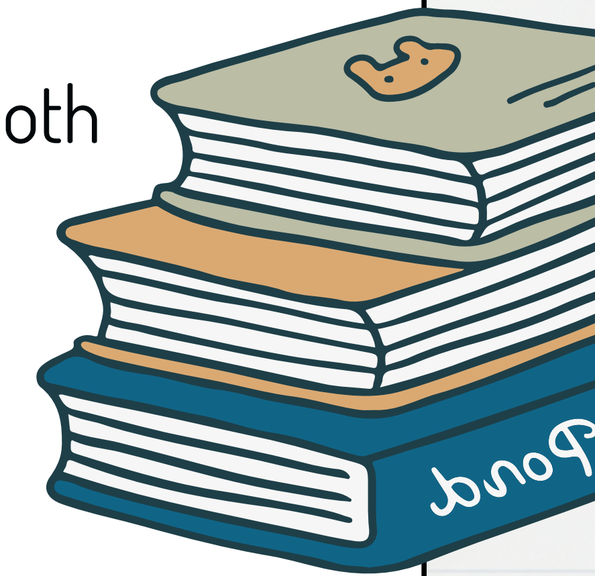
Project Presentation



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Project Overview

The Library Management System project is designed to efficiently manage library resources, members, and transactions through a user-friendly C# application. The system aims to provide a comprehensive solution for library members, ensuring smooth operations and effective resource utilization.

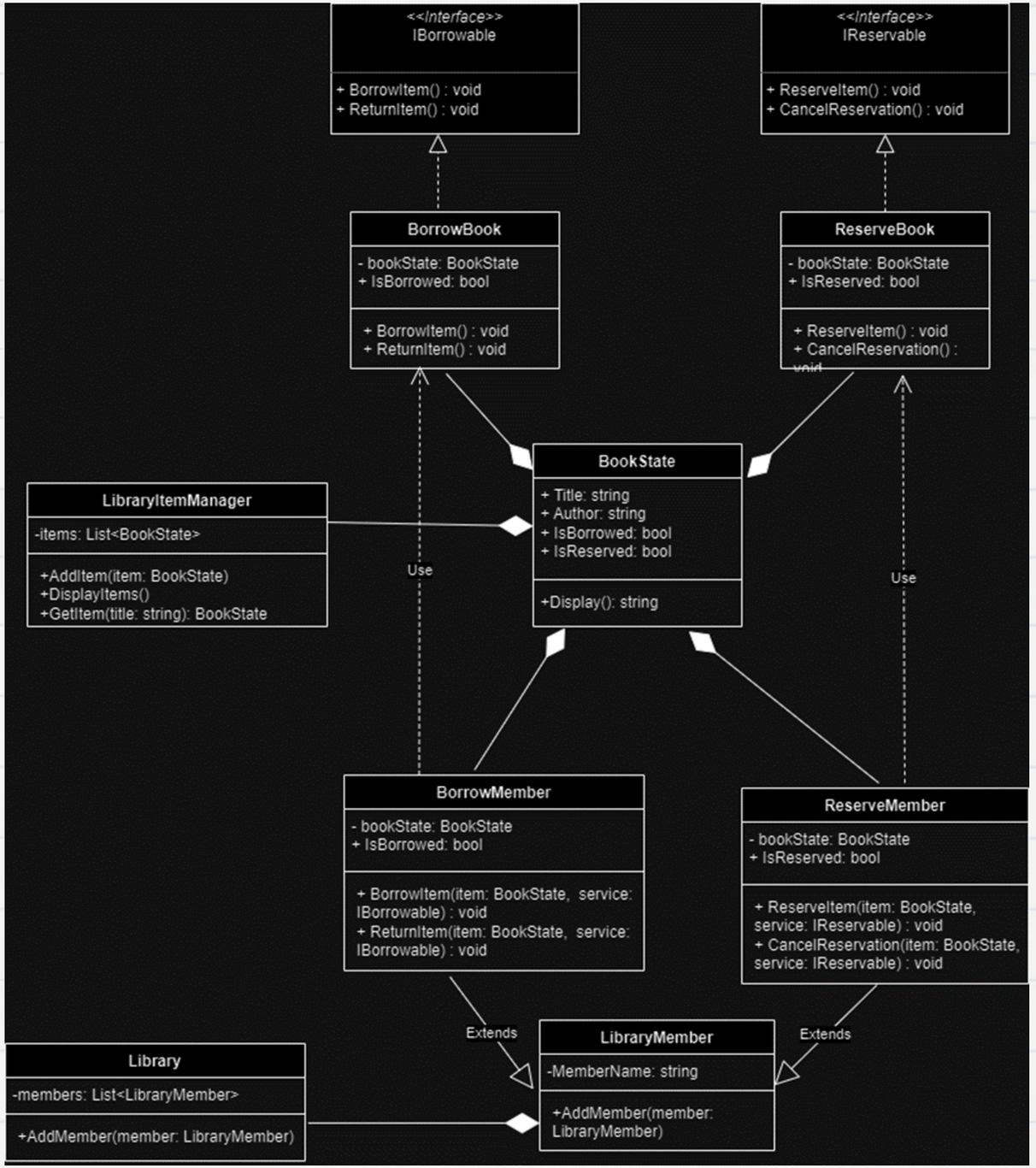


Features

- MANAGE BOOKS
- DISPLAY LIBRARY ITEMS
- MANAGE MEMBERS
- BORROW SYSTEM
- RESERVATION SYSTEM



UML Diagram


<<Interface>>
IBorrowable
+ BorrowItem() : void
+ ReturnItem() : void
<<Interface>>
IReservable
+ ReserveItem() : void
+ CancelReservation() : void
BorrowBook
- bookState: BookState
+ IsBorrowed: bool
+ BorrowItem() : void
+ ReturnItem() : void
ReserveBook
- bookState: BookState
+ IsReserved: bool
+ ReserveItem() : void
+ CancelReservation() : void
BookState
+ Title: string
+ Author: string
+ IsBorrowed: bool
+ IsReserved: bool
+Display(): string
LibraryItemManager
-items: List<BookState>
+AddItem(item: BookState)
+DisplayItems()
+GetItem(title: string): BookState
Use
Use
BorrowMember
- bookState: BookState
+ IsBorrowed: bool
+ BorrowItem(item: BookState, service: IBorrowable) : void
+ ReturnItem(item: BookState, service: IBorrowable) : void
ReserveMember
- bookState: BookState
+ IsReserved: bool
+ ReserveItem(item: BookState, service: IReservable) : void
+ CancelReservation(item: BookState, service: IReservable) : void
Extends
Extends
LibraryMember
-MemberName: string
+AddMember(member: LibraryMember)
Library
-members: List<LibraryMember>
+AddMember(member: LibraryMember)


PLANNER
ACCOUNTS

Following SOLID principles

Single Responsibility Principle

- The BookState class manages the state of a book.
- BorrowBook and ReserveBook handle borrowing and reserving books, respectively .
- LibraryMember, BorrowMember, and ReserveMember are responsible for managing library members.

Open Closed Principle

- IBorrowable and IReservable, allow for easy extension.
- BorrowBook and ReserveBook can be extended for new borrowing or reserving strategies.

Interface Segregation Principle

- IBorrowable and IReservable interfaces have methods specific to borrowing and reserving preventing unnecessary dependencies.

Following SOLID principles

Liskov Substitution Principle

- BorrowMember and ReserveMember (derived classes) can be treated as LibraryMember(base class) .

Dependency Injection Principle

- LibraryItemManager and Library depends on abstractions such as BookState,not on concrete implementations.
- BorrowBook and ReserveBook depend on interfaces IBorrowable and IReservable.



Code smells that are avoided

- LARGE CLASS
- LONG METHOD
- FEATURE ENVY
- DUPLICATED CODE
- INAPPROPRIATE NAMING



Project Demonstration



